

AB-302 & AB-302i Reliability Tests of Adhesion & Visible Transmittance

Everlight Chemical Industrial Corporation, R&D center
2026/06/22

Reliability Test Objectives

Purpose

Evaluate the **adhesion** and **optical degradation** of encapsulant materials when exposed to high-humidity and high-temperature environments.

Encapsulants item

- 1.Eversolar® AB-302
- 2.Eversolar® AB-302i

Test condition

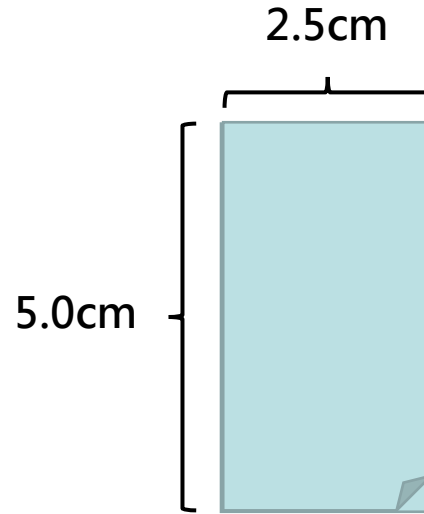
60°C/90%RH@Xhr

Adhesion Comparison of AB-302 & AB-302i (1/2)

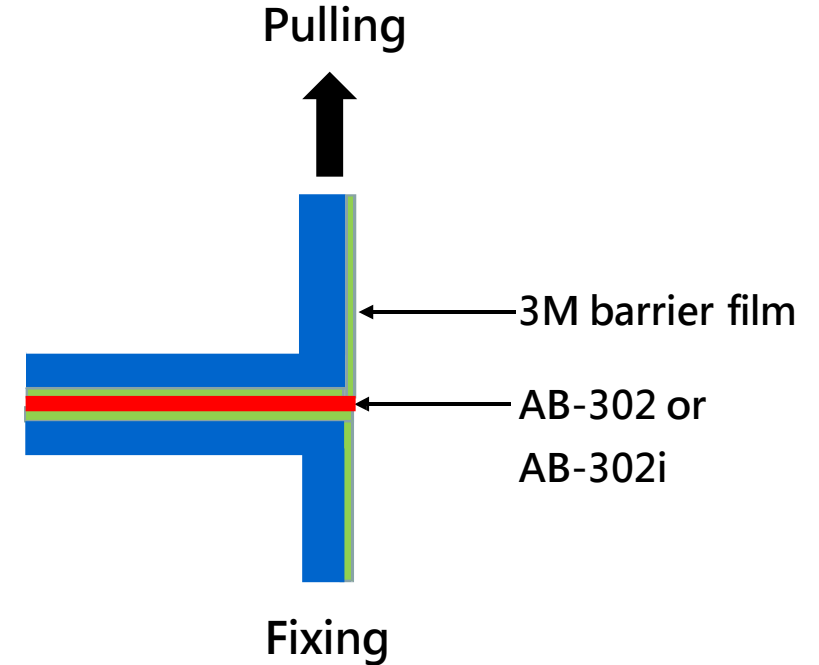
Test method



Instrument used :
1.Environmental chamber
2.Tensile testing machine



Fully laminate two pieces
of 3M barrier film, followed
by UV curing



T-peeling test

Adhesion Comparison of AB-302 & AB-302i (2/2)

Test Result

◆ Aging test condition : 60°C/90%RH@90hr

		AB-302	AB-302i
T-peeling Strength (Film thickness : 100 μ m)	Initial	1.2N/inch	6.5N/inch
	Post-90hr	0.0N/inch	6.4N/inch

- Following aging tests with barrier film lamination, AB-302i demonstrates excellent adhesion and damp heat resistance.

Transmittance Comparison of AB-302 & AB-302i (1/2)

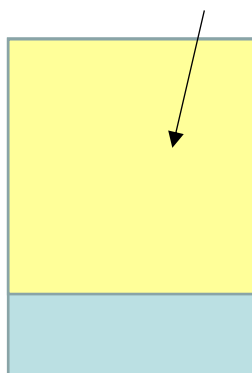
Test method



Instrument used :
1.Environmental chamber
2.Spectrum detective



AB-302 or AB-302i



1.Substrate : Glass
2.Encapsulant thickness :
200 μ m
3.UV curing



Insert the specimen into the
Spectrum detective to determine
the visible light transmittance

Transmittance Comparison of AB-302 & AB-302i (2/2)

Test Result

◆ Aging test condition : 60°C/90%RH@100hr

		AB-302	AB-302i
Transmittance (Film thickness : 200 μ m)	Initial	97%	96%
	Post-100hr	96%	96%

- AB-302 & AB-302i demonstrated excellent reliability in visible light transmittance after damp heat testing.



Thank YOU !